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MaxSketch: Robust Distinct Counting in Streams via Random Projections

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Abstract

Estimating the number of distinct elements in a data stream is well understood when repeated elements are identical. In modern settings, however, observations are high-dimensional and noisy, so repeated instances of the same object are only approximately similar – for example, different images of the same individual may vary significantly at the pixel level. Classical sketches such as HyperLogLog rely on consistent hash values for identical elements and break down in this regime. Recent work on robust distinct counting in general metric spaces achieves $\tilde{\Theta}(\sqrt{n})$ memory, which is tight in the worst case. We show that substantially improved memory guarantees are possible under geometric structure common in learned representations. We introduce MaxSketch, a simple max-linear sketch built from random Gaussian projections, and prove that it succeeds in estimating the number of distinct latent objects. Concretely, we show that under this assumption $m = \tilde{O}(\log n/\varepsilon^2)$ random projections (and hence $\tilde{O}(\log n/\varepsilon^2)$ memory) suffice to recover the true distinct count within a $(1+\varepsilon)$ factor. Experiments on image streams confirm that MaxSketch accurately estimates distinct counts and generalizes beyond the training regime. Our results bridge classical streaming algorithms and modern representation learning, showing how geometric structure can fundamentally reduce the complexity of distinct counting.

1 Introduction

Counting the number of distinct elements in a data stream is a classical algorithmic problem with deep connections to randomized algorithms, information theory, and large-scale data analysis. In its canonical form, the task is to estimate the number of unique keys in a stream using sublinear memory and a single pass over the data. A long line of work has shown that this is possible with striking efficiency: algorithms such as [5, 6, 7] achieve accurate estimates using only logarithmic memory by maintaining carefully designed randomized sketches of the stream.

These classical results, however, rely on a crucial assumption: the same element is represented identically every time it appears. In modern data modalities, this assumption is fundamentally violated. To see this, consider the task of counting the number of distinct people passing through a busy intersection observed by a camera. Suppose Alice and Bob each pass through once, and later Alice passes through again. Although it is the same underlying individual, her second appearance may differ substantially at the pixel level, due to changes in lighting, viewpoint, or pose. As a result, the two observations of Alice may be nearly as different from each other as they are from observations of entirely different individuals.

A natural approach is to embed each photo into a continuous feature space and apply distinct counting in that representation. However, this does not resolve the issue: repeated observations of the same underlying object are no longer identical, causing sketching methods to overcount, while clustering-

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based approaches require maintaining a number of representatives that grows with the number of distinct elements, undermining the sublinear memory guarantees of streaming algorithms.

A recent line of work [2, 3, 30] has studied this *robust* distinct counting problem, where elements are considered equal if they are sufficiently close in some metric. While these results give algorithms with non-trivial guarantees, they also reveal that the problem is fundamentally harder than its classical counterpart: [30] establishes an $\Omega(\sqrt{n})$ memory lower bound in general metric spaces, exponentially far from the rates achievable under exact key equality.

Modern representation learning, however, often induces a more useful geometric structure than the distance-based separation studied above: observations of the same object form tight clusters in embedding space, while different objects correspond to well-separated directions.

This raises a central question:

Closeness of representations in an arbitrary metric space is not strong enough to allow robust distinct counting with polylogarithmic memory ($\Omega(\sqrt{n})$ lower bound in [30]). Does clustered geometry with intra-class closeness and inter-class separation bypass this barrier?

This question sits at the intersection of classical streaming algorithms and modern representation learning. On the one hand, the theory of distinct counting demonstrates that extreme-value statistics of random projections contain sufficient information about the number of unique elements [6, 18]. On the other hand, permutation-invariant architectures such as DeepSets [29] can in principle aggregate set-structured data, but provide no theoretical guarantees on recovering global combinatorial properties like distinct counts. In this work, we bridge this gap with a deliberately minimal max-linear sketch that is both analytically tractable and provably correct.

1.1 A geometric model for distinct objects

We assume that each distinct object corresponds to an unknown latent direction in a high-dimensional space, and that each observation is a noisy but well-aligned version of its object's direction. Different objects occupy well-separated directions, with low inner products between distinct centers. This is not an abstraction: the neural collapse phenomenon formalizes why this configuration emerges in the terminal phase of deep network training [22, 31].

Under this model, the number of distinct objects corresponds to the number of well-separated directions generating the stream. The challenge is to recover this number without explicitly clustering and without storing the embeddings.

The $\Omega(\sqrt{n})$ lower bound of [30] is achieved by adversarial metric spaces with no geometric structure. Our setting is not a special case that sidesteps this lower bound; it is the regime that actually occurs in practice, and one that [30] explicitly leaves open. It is precisely this inner-product structure that enables Gaussian extreme-value statistics whose expectations separate different cardinalities, yielding polylogarithmic memory where the worst-case general-metric setting requires $\tilde{\Omega}(\sqrt{n})$.

1.2 MaxSketch: a max-linear analogue of classical sketches

Our key observation is that max aggregation discards multiplicity by construction: once a cluster has contributed a large projection value, additional repetitions of the same object leave the sketch unchanged. This contrasts sharply with sum or mean aggregation, which scale with frequency and therefore confuse distinctness with multiplicity. Given random Gaussian vectors $w_1, \ldots, w_m \in \mathbb{R}^d$, we compute

$$M_j(X) = \max_{1 \le i \le n} \langle w_j, x_i \rangle$$

where $x_1, \ldots, x_n \in \mathbb{R}^d$ are observations generated from k underlying objects, and average these maxima across projections. The expectation of this statistic grows monotonically with the number of distinct latent directions — the key signal we wish to recover. This operation is permutation-invariant, streaming-compatible, and requires storing only m scalars.

We call this representation MaxSketch. Unlike classical sketches such as HyperLogLog, MaxSketch operates directly on continuous embeddings and tolerates noise and approximate equality. Unlike general-purpose neural set models, MaxSketch is deliberately minimal and analytically tractable.

1.3 Theoretical guarantees

Our main result shows that MaxSketch provably recovers the number of distinct objects up to multiplicative error.

Informal Theorem 4.4 (Robust Distinct Count Estimation). Under a natural clusterability assumption, $m = \tilde{O}(\log n/\varepsilon^2)$ random projections suffice to estimate the number of distinct objects in a stream of length n within a $(1+\varepsilon)$ multiplicative error, with high probability.

Compared to the $\tilde{\Omega}(\sqrt{n})$ memory required in general metric spaces [30]; the improvement is enabled by separation between cluster centers, a property missing from the worst-case instances underlying that bound.

Our analysis combines Gaussian process theory and extreme-value statistics including Slepian's inequality and concentration of Lipschitz functions [1, 20, 26], to show that, despite correlations induced by approximate cluster structure, the maximum projection behaves like the maximum of independent Gaussians up to controlled error terms.

This is a key distinction with other methods. We view MaxSketch as a statistics-preserving embedding: rather than preserving distances or geometry, it maps a set of observations to a scalar statistic that is monotone in the true distinct count. As a result, even when the structural assumptions hold only approximately in practice, the induced statistic remains informative, as illustrated empirically in the face counting experiments of Section 5.4.

Importantly, once the random projections are fixed, the entire dependence of the sketch on the input sequence collapses to a **single scalar statistic** that is monotone in the true number of distinct objects. As a consequence, learning from data reduces to fitting a one-dimensional monotone readout, a task that is sample-efficient and well-conditioned (Theorem 4.7).

1.4 End-to-end learning from count-only supervision

The theoretical guarantees are stated for fixed Gaussian projections, but MaxSketch slots naturally into an end-to-end pipeline: an encoder maps raw inputs to embeddings, MaxSketch aggregates them, and a learned readout produces the count. Our theory does not assume that projections or embeddings are learned optimally; it establishes only that a compact sketch *exists* whenever the embeddings satisfy the geometric assumption. We examine this empirically in Section 5 on MNIST and CIFAR-10, where end-to-end training under count-only supervision induces well-clustered embeddings on simple data, with performance tracking embedding quality as intra-class variability grows.

1.5 Contributions

- **MaxSketch**, a simple max-linear, permutation-invariant sketch inspired by classical extreme-value methods. We prove a $(1+\varepsilon)$-multiplicative guarantee using $\tilde{O}(\log n/\varepsilon^2)$ memory under a natural clusterability assumption improving on the $\tilde{O}(\sqrt{n})$ rate for general metric spaces [30].
- **A learning reduction** showing that, once projections are fixed, count prediction reduces to fitting a one-dimensional monotone function of a fixed scalar statistic.
- **Empirical validation** on MNIST, CIFAR-10, and three merged face datasets, demonstrating extrapolation to sequences longer than those seen at training time and accuracy that tracks embedding geometry rather than sequence length.

1.6 Related Work

Distinct counting and sketching. Estimating the number of distinct elements in a data stream is a classical problem in algorithms, with celebrated solutions such as Flajolet–Martin, LogLog, and HyperLogLog achieving near-optimal accuracy using logarithmic memory via randomized hashing

and extreme-value statistics [5, 6, 7, 18]. These methods crucially rely on exact key equality: repeated occurrences of the same element must hash to the same value. A natural idea in continuous settings is to first apply locality-sensitive hashing (LSH) [16, 17] and then run HyperLogLog on the resulting hash codes. However, this approach fundamentally fails for distinct counting: LSH guarantees collision for some near neighbors, not all repetitions of the same object, and its collision probability depends on distance in a way that entangles object identity with multiplicity. As a result, LSH-based preprocessing introduces bias that cannot be corrected by classical sketches.

Robust distinct counting under near-duplicates. A line of work initiated by [2, 3] studies robust distinct counting under near-duplicates, including well-shaped datasets in low-dimensional Euclidean spaces and, more recently, general metric spaces. Zhang [30] gives $\tilde{O}(\sqrt{n}/\varepsilon)$-space algorithms in general metric spaces and proves an $\Omega(\sqrt{n})$ lower bound for general noisy datasets, characterizing the worst-case cost of robust deduplication without additional geometric structure. Importantly, [30] explicitly leaves open whether the same lower bound holds even for well-shaped inputs.

MaxSketch addresses a different, more structured regime. The stream is embedded in a Euclidean space, observations are tightly aligned with latent centers, and distinct centers have small inner products. This inner-product structure is stronger than a metric distance-separation condition: it allows Gaussian projection maxima to behave as extreme-value statistics whose expectations separate different cardinalities. Under this clusterability assumption, MaxSketch achieves $\tilde{O}(\log n/\varepsilon^2)$ memory, showing that substantially smaller sketches are possible in structured learned representations than in the worst-case general-metric setting.

2 Problem Setting

We observe a stream of unit vectors $X = \langle \tilde{x}_1, \ldots, \tilde{x}_n \rangle \subset \mathbb{S}^{d-1}$, arriving in arbitrary order. Each $\tilde{x}_i$ is a noisy observation of one of $k^\star$ unknown latent objects; the same object may appear multiple times, with each appearance producing a different vector. Our goal is to estimate $k^\star$ in a single pass using memory sublinear in n, with supervision (when available) limited to the total count.

This setting differs from classical distinct-elements counting in that exact equality between repeated occurrences is replaced by approximate geometric similarity. To make the problem well-posed, we assume the stream has the following structure.

Definition 2.1 ($(\eta, \rho)-$clusterable sequence)**.** Fix $\eta \in (0, 1)$ and $\rho \in [0, 1)$. A sequence $X = \langle \tilde{x}_1, \ldots, \tilde{x}_n \rangle$ of unit vectors in $\mathbb{R}^d$ is *(η, ρ)-clusterable* with $k^\star$ centers if there exist distinct unit vectors $x^{(1)}, \ldots, x^{(k^\star)} \in \mathbb{R}^d$ and an assignment $r : [n] \to [k^\star]$ such that:

1. (**Within-cluster alignment**): $\langle \tilde{x}_i, x^{(r(i))} \rangle \geq 1 - \eta/2 \quad$ for all i.

2. (**Cross-cluster separation**): $|\langle x^{(r)}, x^{(s)} \rangle| \leq \rho \quad$ for all $r \neq s$.

The first condition says each observation is directionally close to its center (capturing noise, augmentation, or pose variation); the second says distinct centers are well-separated, with ρ controlling approximate independence under Gaussian projection. This assumption formalizes a structure widely observed in learned representations. When training embeddings for classification, or other self-supervised objectives, the learned space typically exhibits clustering by semantic identity: embeddings of the same object (or class) cluster tightly, while distinct objects occupy well-separated directions. Recent work on neural collapse [10, 22, 31, 32] formalizes this phenomenon in deep networks. Our results do not require learning this geometry, only that it exists.

3 MaxSketch: A Max-Linear Streaming Representation

We now describe a simple sketch that exploits this geometry.

Definition 3.1 (**MaxSketch**)**.** Fix an integer m and let $w_1, \ldots, w_m \sim N(0, I_d)$ be independent random vectors, fixed once and for all. Given a stream $X = \langle x_1, \ldots x_n \rangle$ define

$$M_j(X) = \max_{1 \leq i \leq n} \langle w_j, x_i \rangle, \quad S(X) = \frac{1}{m} \sum_{j=1}^{m} M_j(X).$$

The sketch stores only the m scalar values $M_1, \ldots, M_m$, requiring $O(m)$ memory and supporting single-pass updates.

Why max aggregation? Max aggregation discards multiplicity information by construction: once a cluster has contributed a large projection value, additional repetitions do not change the sketch. This sharply contrasts with sum or mean aggregation, which scale with frequency and therefore confuse distinctness with multiplicity. As we show, under clusterability, maxima of random projections isolate extreme-value statistics that depend primarily on the number of distinct latent directions.

4 Theoretical Guarantees

We show that MaxSketch estimates the number of distinct clusters under $(\eta, \rho)-$clusterability. The analysis has three steps: (i) characterize the extreme-value behavior of Gaussian projections onto well-separated centers, (ii) bound the perturbation introduced by within-cluster noise, and (iii) combine an expectation gap with concentration to distinguish different cluster counts using logarithmically many projections.

4.1 Gaussian maxima over cluster centers

Fix unit vectors $x^{(1)}, \ldots, x^{(k)} \in \mathbb{R}^d$ and $w \sim \mathcal{N}(0, I_d)$, and let $G_r := \langle w, x^{(r)} \rangle$. Then $G = (G_1, \ldots, G_k)$ is jointly Gaussian with mean zero, unit variances, and $\mathrm{Cov}(G_r, G_s) = \langle x^{(r)}, x^{(s)} \rangle$. We control $\mathbb{E}[\max_r G_r]$ when the centers are well-separated by comparing to the i.i.d. case, where extreme-value asymptotics are well understood.

Lemma 4.1 (Gaussian maxima comparison)**.** *Let $G_1, \ldots, G_k$ be mean-zero Gaussians with $\mathrm{Var}(G_r) = 1$ and $|\mathrm{Cov}(G_r, G_s)| \le \rho$, for $r \ne s$, where $\rho \in [0, 1)$ and let $Z_1, \ldots, Z_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$. Then*

$$\sqrt{1-\rho}\,\mathbb{E}\left[\max_{r \le k} Z_r\right] \;\le\; \mathbb{E}\left[\max_{r \le k} G_r\right] \;\le\; \sqrt{1+\rho}\,\mathbb{E}\left[\max_{r \le k} Z_r\right].$$

4.2 Robustness to within-cluster perturbations

Maxima over observed points differ from maxima over centers by an amount controlled by the within-cluster slack η.

Lemma 4.2 (Perturbation bound)**.** *Let $\tilde{X} = \langle \tilde{x}_1, \ldots, \tilde{x}_n \rangle \subset \mathbb{S}^{d-1}$ be (η, ρ)-clusterable with centers $x^{(1)}, \ldots, x^{(k)} \in \mathbb{S}^{d-1}$ and assignment $r : [n] \to [k]$, so that $\langle \tilde{x}_i, x^{(r(i))} \rangle \ge 1 - \eta/2$ for all $i \in [n]$. Let $w \sim \mathcal{N}(0, I_d)$. Then*

$$\left| \mathbb{E}\left[\max_{i \le n} \langle w, \tilde{x}_i \rangle\right] - \mathbb{E}\left[\max_{r \le k} \langle w, x^{(r)} \rangle\right] \right| \;\le\; \sqrt{2\eta \ln n}.$$

4.3 Separating different numbers of clusters

To estimate the number of clusters, we must distinguish between k and $(1+\varepsilon)k$ centers via their expected maxima.

Lemma 4.3 (Gap in Gaussian maxima)**.** *Let $w \sim \mathcal{N}(0, I_d)$ and let $x^{(1)}, \ldots, x^{((1+\varepsilon)k)} \in \mathbb{R}^d$ be unit vectors with $|\langle x^{(i)}, x^{(j)} \rangle| \le \rho$, where $\rho = c_\rho \varepsilon / \ln k$. Define*

$$\Delta_{\mathrm{centers}}(k) := \mathbb{E}\left[\max_{1 \le i \le (1+\varepsilon)k} \langle w, x_i \rangle\right] - \mathbb{E}\left[\max_{1 \le i \le k} \langle w, x_i \rangle\right].$$

Then $\Delta_{\mathrm{centers}}(k) = \Omega(\varepsilon / \sqrt{\ln k})$ for all $k \ge 2$.

To estimate the true k from $S(\tilde{X})$, we must ensure that between-cluster correlations and within-cluster noise does not obscure the signal, and that a $(1+\varepsilon)$ change in cluster count produces a gap large enough to detect. Lemmas 4.1–4.3 establish exactly these three facts, and together they make the main counting theorem possible. Lemmas 4.1–4.3 are proved in Appendix B.1.

4.4 Main counting theorem

Theorem 4.4 (Estimating the number of clusters)**.** *Fix $n \geq 2$, $\varepsilon \in (0, 1/2]$, $\delta \in (0, 1)$. Let $\tilde{X} \subset \mathbb{S}^{d-1}$ be (η, ρ)-clusterable with $k^\star \geq 2$ centers, where $\rho \leq c_\rho \frac{\varepsilon}{\ln n}$ and $\eta \leq c_\eta \frac{\varepsilon^2}{(\ln n)^2}$, for absolute constants $c_\rho, c_\eta > 0$. Draw $w_1, \ldots, w_m \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ with $m \geq C \cdot \dfrac{\ln n \cdot \ln\left(\frac{\ln n}{\varepsilon\delta}\right)}{\varepsilon^2}$ for an absolute constant C. Then there exists an estimator using only*

$$S(\tilde{X}) = \frac{1}{m} \sum_{j=1}^{m} \max_{i \leq n} \langle w_j, \tilde{x}_i \rangle$$

that outputs $\hat{k}$ satisfying $k^\star \leq \hat{k} \leq (1 + \varepsilon)k^\star$, with probability at least $1 - \delta$.

Remark 4.5 (Estimator). A geometric search over the grid $t_r = (1 + \varepsilon)^r$ suffices: each test compares S to a precomputed threshold separating $k \leq t$ from $k \geq (1 + \varepsilon)t$. Importantly, these thresholds are data independent and are derived from the expected maxima of t and $(1 + \varepsilon)t$ i.i.d. Gaussians.

The proof is in Appendix B.2.

4.5 From random to fixed weights, and learning the readout

Theorem 4.4 provides guarantees over a fresh draw of projections for every clusterable input. In practice, however, the sketch is built once and deployed across many sequences from some unknown distribution $\mathcal{D}$. Corollary 4.6 shows that a single fixed sketch suffices for most inputs from $\mathcal{D}$.

Corollary 4.6 (Fixed sketch for a distribution)**.** *Under the conditions of Theorem 4.4, for any distribution $\mathcal{D}$ over $(\eta, \rho)-$clusterable sequences there exist fixed $w_1, \ldots, w_m \in \mathbb{R}^d$ such that MaxSketch achieves $(1 + \varepsilon)$ multiplicative approximation with probability at least $1 - \delta$ over a fresh draw $X \sim \mathcal{D}$.*

Once the projections are fixed, the dependence of the model on X collapses to the scalar statistic $S(X)$, which is monotone in k in expectation. Learning therefore reduces to fitting a one-dimensional monotone readout which is sample-efficient and well-conditioned.

Theorem 4.7 (Learnability)**.** *Under the conditions of Theorem 4.4 there exists a polynomial-time learning algorithm which, given $N = \mathrm{poly}(\ln n, 1/\varepsilon, \ln(1/\delta))$ i.i.d. training samples $(X_1, k_1), \ldots, (X_N, k_N) \sim \mathcal{D}$, outputs a monotone readout function $f : \mathbb{R} \to \mathbb{R}$ such that, for a fresh test sample $(X, k) \sim \mathcal{D}$,$k \leq f(S(X)) \leq (1 + \varepsilon)k$ with probability at least $1 - \delta$.*

Proofs of Corollary 4.6 and Theorem 4.7 are in Appendices B.3–B.4.

Together, Theorem 4.4 and Theorem 4.7 give a complete pipeline: a fixed Gaussian sketch (good with high probability per Theorem 4.4) followed by a monotone readout (efficiently fittable per Theorem 4.7). The next section examines whether this signal remains useful when embeddings themselves are learned end-to-end under aggregate count supervision.

5 Experiments

Our experiments have three goals. First, on MNIST we ask whether an encoder and a MaxSketch readout can be trained jointly from count-only supervision, and whether the resulting model extrapolates to sequence lengths far longer than those seen during training – a direct prediction of the theory once embeddings are clusterable. Second, on CIFAR-10 we ask the same question in a setting with substantially more intra-class variability, and we separate the contribution of the encoder from that of the aggregator by comparing end-to-end training against pretrained features. Third, we evaluate MaxSketch on real face streams using fixed pretrained embeddings, where the assumptions of our theory are only approximately satisfied, and we compare against the streaming algorithm of Zhang [30].

5.1 End-to-end learning to count on MNIST and CIFAR-10

We study the problem of learning to count the number of distinct latent objects in a sequence under aggregate supervision only. Each data point is a sequence $X = (x_1, \ldots, x_n)$ of MNIST/CIFAR-10

images, and the only supervision is the number of distinct latent classes $k(X) = |\{z_1, \ldots, z_n\}|$, appearing in X; the image labels $z_i \in \{1, \ldots, 10\}$ are never observed. The model consists of an encoder $\phi_\theta : \mathcal{X} \to \mathbb{R}^d$, applied independently to each image and an aggregator $f_\psi : (\mathbb{R}^d)^n \to \mathbb{R}$. applied to the resulting set of embeddings, both trained jointly to predict $k(X)$.

Beyond MaxSketch, we include two natural baselines that share the same encoder but differ in f_ψ: a DeepSets aggregator [29], which sum-pools embeddings, and an LSTM aggregator [14] that processes them sequentially. DeepSets is permutation-invariant but conflates frequency with distinctness; the LSTM is included as a non-set baseline.

5.2 MNIST Results

All three models reach near-perfect accuracy across sequence lengths $n = \{2, 5, 10, 20, 50, 100\}$ (Figure 2), and ± 1 accuracy is exactly 100% throughout. This is consistent with the theory: MNIST embeddings have low intra-class variability and strong inter-class separation, so learning an encoder to satisfy the $(\eta, \rho)-$ clusterability assumption becomes substantially easier. In this regime, the aggregation mechanism plays a relatively minor role, and even sum- or sequence-based models can recover the correct count.

Rather than diminishing the contribution of MaxSketch, these results establish an important sanity check: when the clusterability assumption holds strongly, all reasonable aggregation methods succeed, and differences only emerge as the geometry becomes more challenging.

The more informative test is whether the learned model extrapolates beyond the count and length range seen at training. Table 1 reports MaxSketch trained at $n_{\text{train}} \in \{10, 20, 50\}$ and evaluated at $n_{\text{test}} \in \{150, 250, 500\}$. Models trained on shorter sequences exhibit some underestimation bias for very long sequences, but maintain near-perfect ± 1 accuracy, indicating that the learned statistic still carries meaningful count information.

The learned encoder, evaluated independently by 1-NN classification on test images, attains 1.00 accuracy regardless of n_{train}, so the failure at long sequences is not a representation collapse. Rather, it is a calibration failure of the regression head. This behaviour matches the theoretical picture: under clusterability, the MaxSketch statistic is monotone k, so the only learning that needs to happen is fitting a one-dimensional monotone readout, and the difficulty of that fit is determined by the training count range n.

Table 1: Generalization of MaxSketch models to unseen, longer sequence lengths on MNIST. Each model is trained on a specific sequence length n_{train} and evaluated on sequences of lengths n_{test} not seen during training. Numbers indicate exact count accuracy, with ± 1 accuracy shown in parentheses.

n_{train}	$n_{\text{test}} = 150$	$n_{\text{test}} = 250$	$n_{\text{test}} = 500$
10	0.37 (0.97)	0.25 (0.9)	0.06 (0.78)
20	0.70 (0.99)	0.58 (0.98)	0.37 (0.96)
50	0.99 (1.0)	0.98 (1.0)	0.96 (1.0)

5.3 CIFAR-10 Results

CIFAR-10 presents a significantly harder setting, with higher intra-class variability and less pronounced cluster structure and we expect end-to-end training from count supervision alone provides a much weaker learning signal in the representation space. To test this, we compare three regimes that differ only in how the encoder is obtained: **Sup. (Frozen)**, where the encoder is pretrained with class labels and frozen during count training; **Sup. (FT)**, where it is pretrained and then fine-tuned; and **E2E**, where it is trained from scratch using only count supervision. The aggregator is MaxSketch in all three cases. Table 2 shows that pretrained features already support reliable counting and that fine-tuning gives only modest additional gains. End-to-end training from count labels alone is markedly worse, particularly as the sequence length increases. The ± 1 accuracy remains substantial, indicating that even the end-to-end model captures the correct ordering of counts; what it lacks is the cluster structure needed for exact recovery.

These two experiments together draw a clean line: when the encoder produces clusterable embeddings MaxSketch performs as the theory predicts, and the regression head is the only piece that needs to be calibrated for the count range of interest. When the encoder must itself be induced from count supervision in a high-variability regime, performance is dominated by the difficulty of learning clusterable representations from count on information, which lies outside the scope of our analysis.

Table 2: Performance of MaxSketch on CIFAR-10 across sequence lengths, comparing training regimes.

Model	$n_{\text{test}} = 10$	$n_{\text{test}} = 20$	$n_{\text{test}} = 50$
Sup. (Frozen)	0.70 (0.99)	0.69 (0.98)	0.7 (0.97)
Sup. (FT)	0.71 (0.97)	0.72 (0.96)	0.78 (0.99)
E2E	0.55 (0.85)	0.62 (0.83)	0.63 (0.92)

5.4 Counting Unique Faces in Real-World Image Sequences

We now turn to a setting where the embedding is fixed and supplied externally, so that we can isolate MaxSketch from the question of representation learning entirely. We use three face benchmarks: *Labeled Faces in the Wild (LFW)* [15], *CelebA* [21], and the *CMU Multi-PIE* dataset [12], all restricted to identities with at least two images so that each identity can appear multiple times in a stream. The three datasets are merged into a single identity pool, which is split uniformly at random into a calibration half and an evaluation half of $5{,}103$ identities each. For each image we extract a face embedding with the pretrained DeepFace VGG-Face model [24]; embeddings are L2-normalized and held fixed throughout, so the encoder is never adapted to the counting task.

Each evaluation stream is built by drawing a target distinct count $k \in \{100, 200, \ldots, 5100\}$ from the held-out identities and then sampling $n = 20{,}000$ images of those identities, in which the same person typically appears many times. This is the regime motivating our setting in the first place: a video feed or a large photo collection in which repeated observations are the rule. We use $m = 4{,}096$ Gaussian projections and a one-dimensional monotone readout fit on the calibration half by isotonic regression; the readout is fit once and applied unchanged to every evaluation stream. Because k ranges into the thousands, exact recovery is not the right metric, and we instead report the $\pm 50, \pm 100, \pm 150$ accuracy.

The most directly comparable prior work is the streaming algorithm of Zhang for distinct count estimation in general metric spaces. The algorithm is parameterized by a near-duplicate distance threshold α (Definition 1.1 of [30]), and its correctness guarantee assumes the dataset is *well-shaped*: every two points belonging to different underlying objects must be at distance strictly greater than 2α, while every two points belonging to the same object must be at distance at most α. Real face embeddings do not satisfy this condition at any α as intra-identity and inter-identity distance distributions overlap. One should therefore expect the algorithm to be sensitive to the choice of α and to errors of both signs. We implement Algorithm 1 of [30] faithfully and select $\alpha = 0.40$ as the value that minimizes error on the calibration half (the same split used to fit our readout). Across the full evaluation range $k \in \{100, 200, \ldots, 5100\}$, Zhang's algorithm fails to achieve ± 150 accuracy on 43 of the 51 evaluated values of k.

By contrast, due to MaxSketch's nature as a "statistical embedding" mild violations of $(\eta, \rho)-$clusterability shift the expectation of $S(X)$ continuously, and the calibrated readout absorbs these biases. Empirically (Figure 1), MaxSketch maintains at least 60% accuracy at the strictest ± 50 tolerance across the full k range using a single readout, and the ± 100 and ± 150 curves remain near 100% throughout. Overall, this demonstrates that MaxSketch can extract coarse multiplicity information directly from pretrained embedding spaces without requiring clustering or identity supervision at inference time. Such approximate counts may be useful in large-scale monitoring and analytics scenarios where rough population estimates are sufficient and explicit identity resolution is unnecessary.

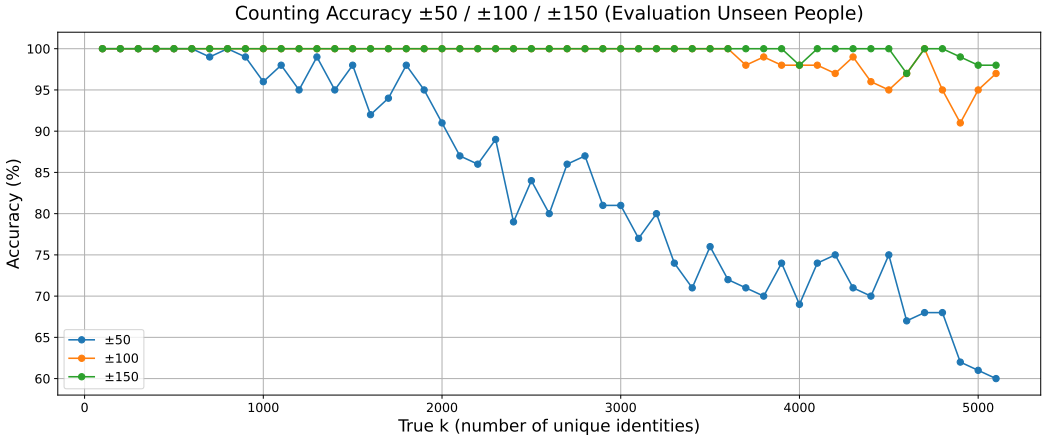


Figure 1: MaxSketch counting accuracy on the held-out face identity split, reported at tolerances of ± 50, ± 100, and ± 150 across $k \in \{100, 200, \ldots, 5100\}$.

6 Limitations

Our theoretical guarantees rely on an explicit geometric separation assumption: embeddings must be $(\eta, \rho)-$clusterable, with small within-cluster variance and sufficiently weak correlations across cluster centers. When this assumption is violated, e.g., in highly entangled visual data with substantial intra-class variability or semantically overlapping categories, the gap in expected maxima necessarily diminishes, making reliable separation of cluster counts information-theoretically difficult. The CIFAR-10 and LFW experiments illustrate this regime: while MaxSketch continues to capture coarse multiplicity information, exact counting degrades as embeddings deviate from well-separated cluster structure.

Additionally, understanding when count supervision alone can provably induce clusterable embedding geometry, rather than merely exploit it when present, remains a central open problem.

7 Conclusion

We introduced MaxSketch, a max-linear streaming sketch built from random Gaussian projections, and showed that it provably estimates the number of distinct latent objects in a stream under a natural clusterability assumption. The central insight is that geometric structure common in modern learned representations, that is tight within-cluster alignment and weak cross-cluster correlation, is precisely the structure needed to make extreme-value statistics informative about cardinality and surpass the known lower bound. Under this assumption, MaxSketch achieves $(1 + \varepsilon)-$multiplicative accuracy using only $\tilde{O}(\log n/\varepsilon^2)$ memory.

On the practical side, MaxSketch achieves high accuracy when embeddings are well-clustered, as in MNIST or pretrained CIFAR features, but its exact accuracy decreases on datasets with substantial intra-class variability, such as end-to-end CIFAR. On real face streams, where clusterability holds only approximately, the calibrated readout absorbs residual bias from mild violations, yielding robust approximate counts across a wide identity range.

Two open problems would sharpen the theoretical picture. First, no lower bound is currently known under our clusterability assumption; establishing one would clarify whether MaxSketch's memory is optimal in the structured regime. Second, our guarantees assume clusterability is already present in the encoder; a theory of when aggregate count supervision alone can provably induce this geometry would bring the theoretical and empirical settings into alignment, replacing a geometric assumption on the encoder with a condition on the learning objective alone.

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A Appendix: Extended Related Work

A.1 Classical Sketching and Cardinality Estimation

The problem of estimating the number of distinct elements in a stream has been extensively studied in the data management and streaming algorithms literature. Flajolet and Martin [7] pioneered probabilistic counting via random bit patterns, achieving $O(\log^2 n)$ space for $O(1/\varepsilon^2)$ multiplicative approximation. Durand and Flajolet [5] improved this via the LogLog algorithm, reducing space to $O(\log n)$ registers. The authors in [13] introduced HyperLogLog, which achieves $O(\log^2 n)$ space with $(1+\varepsilon)$ multiplicative approximation and remains the gold standard in classical sketching, widely used in databases (e.g., Redis, Presto) and big data systems.

These classical methods share two defining features: (i) they assume deterministic, identical representations for repeated elements (via cryptographic hashing), and (ii) they use hand-designed random projections and analytical estimators derived from probabilistic analysis. While highly effective, this paradigm raises a natural question: can modern machine learning discover task-specific sketching statistics end-to-end?

A.2 Permutation-Invariant Models and Set Functions

Recent work has clarified the theory of permutation-invariant neural networks operating on sets. The authors in [29] established that all permutation-invariant functions on finite sets have the form $\rho(\sum \phi(x_i))$, providing a complete characterization. This DeepSets framework has become foundational for learning on variable-sized sets and inspired numerous extensions.

A.3 Self-Supervised Learning and Representation Quality

Our method's success depends critically on good embeddings satisfying clusterability (Definition 2.1). Recent breakthroughs in self-supervised learning have produced embeddings where similar items cluster and dissimilar items separate, directly supporting our geometric assumptions.

SimCLR [4]: Uses contrastive loss to learn representations where augmentations of the same image cluster, and different images separate. Achieves clustering structure aligned with our geometric model, producing embeddings with strong within-class alignment and between-class separation.

MoCo [11]: Momentum contrast provides an alternative contrastive framework, showing robust clustering across diverse datasets without requiring large batch sizes. Demonstrates the robustness of clustering assumptions across different SSL architectures.

BYOL [9]: Learns clustering without explicit negative pairs, showing that clustering emerges naturally even without contrastive objectives. Further validates that learned representations naturally satisfy clusterability.

Neural Collapse Phenomenon [22, 31]: Recent work formalizes why deep classifiers learn near-orthogonal class representations in the terminal phase of training. [32] provide further geometric characterization, confirming that learned embeddings have the low-correlation structure we assume.

A.4 Counting Objects in Images

[8] is a survey about direct count regression in images (e.g., counting people in a crowd). These methods differ fundamentally from our work: they predict total counts in a single image, not distinct elements in a stream. No theoretical guarantees or handling of weak supervision.

A.5 Permutation-invariant models and max aggregation.

In deep learning, permutation-invariant models such as DeepSets aggregate unordered inputs via sum, mean, or max pooling and are universal approximators of set functions under sufficient capacity [29]. However, the choice of aggregator has a decisive impact on what statistics can be recovered: sum and mean aggregates conflate frequency and distinctness, and fixed-dimensional sum-based representations impose fundamental limitations on the expressivity of permutation-invariant set functions under continuous mappings [23, 27, 28]. Recent work has explored learnable aggregation

mechanisms, showing that approximating complex set functions often requires aggregators beyond fixed sum or max and can incur poor learnability or require latent dimensions scaling with set size [19, 23]. In contrast, our work provides finite-sample, multiplicative error guarantees for a specific max-linear sketch under a transparent geometric clusterability assumption, explaining why max projections can isolate extreme-value statistics that correlate monotonically with distinct counts.

B Proofs of Theorems 1–3

B.1 Preliminaries and technical lemmas

Throughout, for an integer $k \geq 1$, define

$$G_k \;:=\; \langle w, x_k \rangle, \qquad w \sim \mathcal{N}(0,1), \qquad |\langle x_i, x_j \rangle| \leq \rho$$

Max sketch statistic. Given a sequence $\tilde{X} = \langle \tilde{x}_1, \ldots, \tilde{x}_n \rangle \subset \mathbb{S}^{d-1}$ and weights $w_1, \ldots, w_m \in \mathbb{R}^d$, define

$$M_j(\tilde{X}) \;:=\; \max_{1 \leq i \leq n} \langle w_j, \tilde{x}_i \rangle, \qquad S(\tilde{X}) \;:=\; \frac{1}{m} \sum_{j=1}^{m} M_j(\tilde{X}).$$

Lemma B.1 (Norm bound for Gaussians)**.** *If $w \sim \mathcal{N}(0, I_d)$ then $\mathbb{E}\|w\| \leq \sqrt{d}$ and $\Pr(\|w\| \geq \sqrt{d} + t) \leq e^{-t^2/2}$.*

Proof. The tail bound is standard for χ^2_d concentration (e.g. by Gaussian concentration for the 1-Lipschitz function $w \mapsto \|w\|$). Integrating the tail (or Jensen) yields $\mathbb{E}\|w\| \leq \sqrt{\mathbb{E}\|w\|^2} = \sqrt{d}$. □

Lemma B.2 (Perturbation bound)**.** *Let $\tilde{X} = \langle \tilde{x}_1, \ldots, \tilde{x}_n \rangle \subset \mathbb{S}^{d-1}$ be (η, ρ)-clusterable with centers $x^{(1)}, \ldots, x^{(k)} \in \mathbb{S}^{d-1}$ and assignment $r : [n] \to [k]$, so that $\langle \tilde{x}_i, x^{(r(i))} \rangle \geq 1 - \eta/2$ for all $i \in [n]$. Let $w \sim \mathcal{N}(0, I_d)$. Then*

$$\Big| \, \mathbb{E}\Big[\max_{i \leq n} \langle w, \tilde{x}_i \rangle \Big] - \mathbb{E}\Big[\max_{r \leq k} \langle w, x^{(r)} \rangle \Big] \, \Big| \;\leq\; \sqrt{2\eta \ln n}.$$

Proof. For each i, define $v_i := \tilde{x}_i - x^{(r(i))}$. By clusterability, $\|v_i\|^2 \leq \eta$. Then $\langle w, v_i \rangle \sim \mathcal{N}(0, \|v_i\|^2) \subseteq \mathcal{N}(0, \eta)$.

Next, using $|\max_i a_i - \max_i b_i| \leq \max_i |a_i - b_i|$ and linearity of expectation:

$$\Big| \mathbb{E}[\max_i \langle w, \tilde{x}_i \rangle] - \mathbb{E}[\max_i \langle w, x^{(r(i))} \rangle] \Big| \;\leq\; \mathbb{E}\Big[\max_i |\langle w, v_i \rangle| \Big] \;\leq\; \sqrt{2\eta \log n}.$$

□

Lemma B.3 (Gaussian maxima comparison (Slepian/Sudakov–Fernique style))**.** *Let $G_1, \ldots, G_k$ be mean-zero Gaussians with Var$(G_r) = 1$ and $|Cov(G_r, G_s)| \leq \rho$ for $r \neq s$. Let $Z_1, \ldots, Z_k \overset{i.i.d.}{\sim} \mathcal{N}(0,1)$. Then*

$$\sqrt{1-\rho}\, \mathbb{E}\Big[\max_{r \leq k} Z_r \Big] \;\leq\; \mathbb{E}\Big[\max_{r \leq k} G_r \Big] \;\leq\; \sqrt{1+\rho} \mathbb{E}\Big[\max_{r \leq k} Z_r \Big].$$

Proof. We will use the Sudakov-Fernique inequality which says that if

$$\mathbb{E}\big[(X_i - X_j)^2\big] \;\leq\; \mathbb{E}\big[(Y_i - Y_j)^2\big], \quad \text{for all } i, j = 1, \ldots, n. \tag{1}$$

Then

$$\mathbb{E}\Big[\max_{i=1,\ldots,n} X_i \Big] \;\leq\; \mathbb{E}\Big[\max_{i=1,\ldots,n} Y_i \Big]. \tag{2}$$

Let $Z \sim \mathcal{N}(0,1)$ be independent of $Z_1, \ldots, Z_k$ and define

$$G'_r \;:=\; \sqrt{1-\rho}\, Z_r + \sqrt{\rho}\, Z.$$

Then each $G'_r \sim \mathcal{N}(0,1)$ and $\mathrm{Cov}(G'_r, G'_s) = \rho$ for $r \neq s$. By Sudakov Fernique inequality, since $\mathbb{E}[G'^2_r] + \mathbb{E}[G'^2_s] - 2\mathbb{E}[G'_r G'_s] \leq \mathbb{E}[G^2_r] + \mathbb{E}[G^2_s] - 2\mathbb{E}[G_r G_s]$ for $r \neq s$ with equal variances,

$$\mathbb{E}\Big[\max_{r\leq k} G_r\Big] \;\geq\; \mathbb{E}\Big[\max_{r\leq k} G'_r\Big].$$

But $\max_r G'_r = \sqrt{1-\rho}\max_r Z_r + \sqrt{\rho}\, Z$, so taking expectation and using $\mathbb{E}[Z] = 0$ yields

$$\mathbb{E}\Big[\max_{r\leq k} G_r\Big] \;\geq\; \sqrt{1-\rho}\,\mathbb{E}\Big[\max_{r\leq k} Z_r\Big].$$

For the upper bound, we apply Sudakov Fernique inequality using $Y_i = \sqrt{1+\rho}Z_i$. For this random variable we have $\mathbb{E}[Y_i] = 0$, $\mathrm{Var}[Y_i] = 1+\rho$ and $\mathrm{Cov}(Y_i Y_j) = 0$. This means that $\mathbb{E}[Y_i^2] + \mathbb{E}[Y_j^2] - 2\mathbb{E}[Y_i Y_j] = 2 + 2\rho \geq \mathbb{E}[G_r^2] + \mathbb{E}[G_s^2] - 2\mathbb{E}[G_r G_s]$. Since $\mathbb{E}[\max_{r\leq k} Y_i] = \sqrt{1+\rho}\mathbb{E}[\max_{r\leq k} Z_i]$, we get

$$\sqrt{1+\rho}\mathbb{E}[\max_{r\leq k} Z_i] \;\geq\; \mathbb{E}\Big[\max_{r\leq k} G_r\Big].$$

□

Lemma B.4 (Concentration of Gaussian max projections)**.** *Fix unit vectors $u_1, \ldots, u_n \in \mathbb{S}^{d-1}$ and define $f(w) = \max_{i\leq n}\langle w, u_i\rangle$. Then f is 1-Lipschitz: $|f(w) - f(w')| \leq \|w - w'\|$. Consequently, for $w \sim \mathcal{N}(0, I_d)$,*

$$\Pr\Big(|f(w) - \mathbb{E}f(w)| \geq t\Big) \;\leq\; 2e^{-t^2/2}.$$

Moreover, if $w_1, \ldots, w_m \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ and $F := \frac{1}{m}\sum_{j=1}^m f(w_j)$, then

$$\Pr\Big(|F - \mathbb{E}F| \geq t\Big) \;\leq\; 2\exp\Big(-\frac{mt^2}{2}\Big).$$

Proof. For any w, w',

$$f(w) - f(w') \;=\; \max_i\langle w, u_i\rangle - \max_i\langle w', u_i\rangle \;\leq\; \max_i\langle w - w', u_i\rangle \;\leq\; \|w - w'\|,$$

and similarly with w, w' swapped, so f is 1-Lipschitz. Gaussian concentration for Lipschitz functions yields the tail. For the average, apply independence and subgaussianity: each $f(w_j) - \mathbb{E}f(w_j)$ is 1-subgaussian, hence the average is $1/\sqrt{m}$-subgaussian. □

Lemma B.5 (Mean gap)**.** *Let $Z_1, \ldots, Z_{k'} \overset{\text{i.i.d.}}{\sim} N(0,1)$ with $k' = \lceil(1+\varepsilon)k\rceil$. Define*

$$\Delta_1(k, \varepsilon) := \mathbb{E}\left[\max_{i\leq k'} Z_i\right] - \mathbb{E}\left[\max_{i\leq k} Z_i\right].$$

There exists an absolute constant $c > 0$ such that for all integers $k \geq 2$ and all $\varepsilon \in (0, 1]$,

$$\Delta_1(k, \varepsilon) \;\geq\; \frac{c\,\varepsilon}{\sqrt{\ln k}}.$$

Proof. Let

$$M := \max_{i\leq k} Z_i, \qquad N := \max_{k<i\leq k'} Z_i, \qquad m := k' - k.$$

Since the index sets are disjoint and the Z_i are independent, M and N are independent. Also $m = \lceil(1+\varepsilon)k\rceil - k \geq \varepsilon k$, with $m \geq 1$ whenever $\varepsilon k > 0$.

We first show a pointwise inequality. For every realization and every $t \in \mathbb{R}$,

$$(N - M)_+ \;\geq\; (N - t)_+ \cdot \mathbf{1}_{\{M\leq t\}}.$$

This follows by a simple case analysis:

- If $M \leq t$ and $N \leq t$: LHS $= (N - M)_+ \geq 0$, RHS $= 0$. OK.

- If $M \le t$ and $N > t$: LHS $= N - M$ (since $N > t \ge M$ implies $N > M$), RHS $= N - t$. We need $N - M \ge N - t$, equivalently $M \le t$, which holds.
- If $M > t$ and $N \le t$: RHS $= 0$ (indicator vanishes); LHS ≥ 0. OK.
- If $M > t$ and $N > t$: RHS $= 0$; LHS ≥ 0. OK.

Taking expectations and using the independence of M and N:

$$\Delta_1(k, \varepsilon) = \mathbb{E}[\max_{i \le k'} Z_i - \max_{i \le k} Z_i] \overset{(*)}{=} \mathbb{E}[(N - M)_+] \ge \Pr(M \le t) \cdot \mathbb{E}[(N - t)_+]. \tag{3}$$

The equality $(*)$ uses that $\max_{i \le k'} Z_i = \max(M, N)$, and $\max(M, N) - M = (N - M)_+$.

Set

$$t := \Phi^{-1}\big(1 - \tfrac{1}{2k}\big), \qquad \text{so} \qquad 1 - \Phi(t) = \tfrac{1}{2k}.$$

We will lower bound $\mathbb{E}[(N - t)_+]$ and combine with (3).

First, we prove that $\Pr(M \le t) \ge \frac{1}{2}$ for all $k \ge 1$.

$$\Pr(M \le t) = \Phi(t)^k = \big(1 - \tfrac{1}{2k}\big)^k.$$

Since $(1 - \frac{1}{2k})^k$ is increasing in $k \ge 1$, we have

$$\Pr(M \le t) = \left(1 - \frac{1}{2k}\right)^k \ge \frac{1}{2}. \tag{4}$$

Additionally, we need to prove an upper bound on t, for all $k \ge 2$.

The standard Gaussian tail bound (Chernoff) gives $1 - \Phi(t) \le e^{-t^2/2}$ for $t \ge 0$. Thus $1/(2k) = 1 - \Phi(t) \le e^{-t^2/2}$, giving $t^2/2 \le \ln(2k)$, i.e., $t \le \sqrt{2\ln(2k)}$. For $k \ge 2$, $2k \le k^2$, so $\ln(2k) \le 2\ln k$, giving $t \le 2\sqrt{\ln k}$.

Furthermore, we need to prove a bound on $t + 2$. We see that the ratio $(2\sqrt{\ln k} + 2)/(5\sqrt{\ln k})$ is decreasing in k, so the bound

$$t + 2 \le 5\sqrt{\ln k} \tag{5}$$

holds for all $k \ge 2$ We now derive a uniform lower bound on $\Pr(N > t + s)$ for small s, which will yield a lower bound on $\mathbb{E}[(N - t)_+$ via integration.

With t as defined and $\delta := 1/(2(t + 2))$, for all $\varepsilon \in (0, 1]$ and $k \ge 2$,

$$\mathbb{E}[(N - t)_+] \ge \frac{c_0\, \varepsilon}{2(t + 2)}, \qquad c_0 := \frac{1 - e^{-1}}{2e^2}.$$

We now control $\Pr(N > t + s)$ uniformly for $s \in [0, \delta]$ and then integrate.

(a) Hazard-rate bound. The Gaussian hazard rate is $h(u) := \phi(u)/(1 - \Phi(u))$ for $u \in \mathbb{R}$. We claim:

$$h(u) \le 2(u + 1) \quad \text{for all } u \ge 0. \tag{6}$$

For $u \ge 1$: by Mills' ratio, $1 - \Phi(u) \ge \frac{u}{1+u^2}\phi(u)$, so $h(u) \le \frac{1+u^2}{u} = u + \frac{1}{u} \le u + 1 \le 2(u + 1)$.

A direct computation shows $h(1) \approx 1.525$, and $h(u)$ is increasing on $[0, \infty)$, hence $h(u) \le h(1)$ for $u \in [0, 1]$.

(b) Integral bound. For $s \in [0, \delta]$, by (6),

$$\int_t^{t+s} h(u)\, du \le 2 \int_t^{t+s} (u + 1)\, du = 2s(t + 1) + s^2 \le 2\delta(t + 1) + \delta^2.$$

Substituting $\delta = 1/(2(t + 2))$:

$$2\delta(t + 1) + \delta^2 = \frac{t + 1}{t + 2} + \frac{1}{4(t + 2)^2} \le 1 + \frac{1}{4(t + 2)^2} \le 1 + \frac{1}{16} \le 2,$$

where we used $t \geq 0$, hence $t + 2 \geq 2$ and $1/(4(t+2)^2) \leq 1/16$.

(c) Tail of Φ. The fundamental theorem of calculus applied to $-\ln(1-\Phi)$ (whose derivative is h) gives

$$1 - \Phi(t+s) = (1-\Phi(t)) \cdot \exp\left(-\int_t^{t+s} h(u)\,du\right) \geq \frac{1}{2k} \cdot e^{-2} \quad \text{for all } s \in [0, \delta].$$

(d) Tail of N. Since $N = \max_{k<i\leq k'} Z_i$ is the max of m i.i.d. standard normals,

$$\Pr(N > t+s) = 1 - \Phi(t+s)^m \geq 1 - \exp(-m(1-\Phi(t+s))),$$

where we used $(1-p)^m \leq e^{-mp}$ for $p \in [0,1]$. Combining with (c) and $m \geq \varepsilon k$:

$$m(1-\Phi(t+s)) \geq \varepsilon k \cdot \frac{e^{-2}}{2k} = \frac{\varepsilon\, e^{-2}}{2}.$$

Setting $x := \varepsilon e^{-2}/2 \in (0, e^{-2}/2] \subset (0,1]$, we use the elementary inequality $1 - e^{-x} \geq (1-e^{-1})x$ for $x \in [0,1]$ (the function $g(x) = (1-e^{-x})/x$ is decreasing on $(0,\infty)$ with $g(1) = 1 - e^{-1}$):

$$\Pr(N > t+s) \geq 1 - e^{-x} \geq (1-e^{-1})x = (1-e^{-1}) \cdot \frac{\varepsilon\, e^{-2}}{2} = c_0\, \varepsilon.$$

(e) Integration. Using $\mathbb{E}[(N-t)_+] = \int_0^\infty \Pr(N > t+s)\,ds$:

$$\mathbb{E}[(N-t)_+] \geq \int_0^\delta c_0\, \varepsilon\, ds = c_0\, \varepsilon \cdot \delta = \frac{c_0\, \varepsilon}{2(t+2)}. \tag{7}$$

□

Step 6: Combining the bounds

By Equations(3), (4), (5), and (7):

$$\Delta_1(k,\varepsilon) \geq \Pr(M \leq t) \cdot \mathbb{E}[(N-t)_+] \geq \frac{1}{2} \cdot \frac{c_0\, \varepsilon}{2(t+2)} \geq \frac{c_0\, \varepsilon}{4 \cdot 5\sqrt{\ln k}} = \frac{c_0\, \varepsilon}{20\sqrt{\ln k}}.$$

which proves the claim. □

Lemma B.6 (A quantitative gap for i.i.d. correlated Gaussian maxima)**.** *Let $w \sim \mathcal{N}(0, I_d)$, and let $x^{(1)}, \ldots, x^{((1+\varepsilon)k)} \in \mathbb{R}^d$, be unit norm vectors satisfying $|\langle x^{(i)}, x^{(j)}\rangle| \leq \rho$, with $\rho = c_\rho \frac{\varepsilon}{\ln k}$. Define*

$$\Delta_{\textit{centers}}(k) = \mathbb{E}\left[\max_{1\leq i\leq (1+\varepsilon)k} \langle w, x_i\rangle\right] - \mathbb{E}\left[\max_{1\leq i\leq k} \langle w, x_i\rangle\right]$$

Then we have:

$$\Delta_{\textit{centers}}(t) = \Omega\left(\frac{\varepsilon}{\sqrt{\log k}}\right) \quad \textit{for all } k \geq 2$$

Proof. We have that $G_i = \langle w, x_i\rangle$ is a mean-zero Gaussian with $\mathrm{Var}(G_i) = 1$. Additionally, since we have that $\langle x^{(r)}, x^{(s)}\rangle \leq \rho$, we get for $r \neq s$,

$$\mathrm{Cov}(G_r,\, G_s) \;=\; \mathbb{E}\big[G_r\, G_s\big] \;=\; \big\langle x^{(r)},\, x^{(s)}\big\rangle \;\leq\; \rho.$$

Therefore, from Lemma B.3, we get that lower bounding $\Delta_{\text{centers}}(k)$ reduces to lower bounding

$$\Delta(k) := \sqrt{1-\rho}\,\mathbb{E}\left[\max_{1\leq i\leq (1+\varepsilon)k} Z_i\right] - \sqrt{1+\rho}\mathbb{E}\left[\max_{1\leq i\leq k} Z_i\right],$$

where $Z_1, \ldots, Z_k \sim N(0,1)$ i.i.d.

We rearrange this to get:

$$\Delta(k) = \underbrace{\mathbb{E}\left[\max_{1\le i\le (1+\varepsilon)k} Z_i\right] - \mathbb{E}\left[\max_{1\le i\le k} Z_i\right]}_{\Delta_1} - (1-\sqrt{1-\rho})\underbrace{\mathbb{E}\left[\max_{1\le i\le (1+\varepsilon)k} Z_i\right]}_{\Delta_2} - (\sqrt{1+\rho}-1)\underbrace{\mathbb{E}\left[\max_{1\le i\le k} Z_i\right]}_{\Delta_3}$$

We focus on $\Delta_1 := \mathbb{E}\left[\max_{1\le i\le (1+\varepsilon)k} Z_i\right] - \mathbb{E}\left[\max_{1\le i\le k} Z_i\right]$.

Write $k' = \lceil (1+\varepsilon)k \rceil$. Let $q(\alpha)$ denote the upper α-tail quantile of a standard normal: $1-\Phi(q(\alpha)) = \alpha$, so in particular $\Pr(Z \ge q(\alpha)) = \alpha$.

By Lemma B.5 we get

$$\Delta_1 \;\ge\; c\,\frac{\varepsilon}{\sqrt{\ln k}},$$

Then, for Δ_2 and Δ_3 we have that,

$$(1-\sqrt{1-\rho})\mathbb{E}\left[\max_{1\le i\le k'} Z_i\right] \le \rho\sqrt{2\ln k'}$$

and

$$(\sqrt{1+\rho}-1)\mathbb{E}\left[\max_{1\le i\le k} Z_i\right] \le \rho\sqrt{2\ln k}$$

because $\rho \in [0,1]$ and a typical upper bound on the expected maximum of independent Gaussians. Choosing $\rho = c_\rho\left(\frac{\varepsilon}{\ln k}\right)$, for sufficiently small constant c_ρ we keep the claimed lower bound. □

B.2 Theorem 1: Random max-aggregator counts any clusterable sequence

Theorem B.7 (Theorem 4.4 (fully explicit statement))**.** *Fix $n \ge 2$, $\varepsilon \in (0, 1/2]$, and $\delta \in (0,1)$. Let $\tilde{X} = \langle \tilde{x}_1, \dots, \tilde{x}_n \rangle \subset \mathbb{S}^{d-1}$ be (η,ρ)-clusterable with $k^\star$ centers. Assume*

$$\rho \;\le\; c_\rho \frac{\varepsilon}{\ln n}, \qquad \eta \;\le\; c_\eta \frac{\varepsilon^2}{(\ln n)^2},$$

for sufficiently small absolute constants $c_\rho, c_\eta > 0$.

Draw $w_1, \dots, w_m \overset{i.i.d.}{\sim} \mathcal{N}(0, I_d)$ with

$$m \;\ge\; C \cdot \frac{\ln n \cdot \ln\left(\frac{\ln n}{\varepsilon\,\delta}\right)}{\varepsilon^2}$$

for a sufficiently large absolute constant C.

Then there is an explicit estimator (described below) using only the statistic $S(\tilde{X}) = \frac{1}{m}\sum_{j=1}^{m} \max_{i\le n}\langle w_j, \tilde{x}_i\rangle$ that outputs $\hat{k}$ satisfying

$$k^\star \;\le\; \hat{k} \;\le\; (1+\varepsilon)\,k^\star$$

with probability at least $1-\delta$ over the draw of $w_1, \dots, w_m$.

Estimator (geometric search). Let $t_0 = 2$ and $t_r = \lceil (1+\varepsilon)^r \rceil$ for $r = 0, 1, \dots, R$, where R is the smallest index with $t_R \ge n$. For each $r < R$, define the deterministic threshold

$$\theta_r \;:=\; \frac{1}{2}\Big(U(t_r) + L(t_{r+1})\Big)$$

where

$$U(t) = \sqrt{1+\rho}\,\mathbb{E}\Big[\max_{r\le t} Z_r\Big] + \sqrt{2\eta\log n}, \quad L(t) = \sqrt{1-\rho}\,\mathbb{E}\Big[\max_{r\le t} Z_r\Big] - \sqrt{2\eta\log n}.$$

Compute $S = S(\tilde{X})$ and return the first index r such that $S \le \theta_r$, outputting $\hat{k} = t_{r+1}$. (If no such r occurs, output $\hat{k} = t_R = n$.)

Proof of Theorem B.7. The proof has three parts: (i) bounding $\mathbb{E}[S]$ for a fixed sequence, (ii) exhibiting a uniform gap between $k^\star \le t$ and $k^\star \ge (1+\varepsilon)t$, and (iii) concentration plus a union bound over the geometric grid.

Step 1: Expectation bounds via centers and perturbations. Let the $k^\star$ distinct centers be $x^{(1)}, \dots, x^{(k^\star)} \in \mathbb{S}^{d-1}$. For a single $w \sim \mathcal{N}(0, I_d)$ define

$$M(\tilde{X}; w) := \max_{i \le n} \langle w, \tilde{x}_i \rangle, \qquad M(T; w) := \max_{r \le k^\star} \langle w, x^{(r)} \rangle.$$

Lemma B.2,

$$\Big| \mathbb{E}_w M(\tilde{X}; w) - \mathbb{E}_w M(T; w) \Big| \;\le\; \sqrt{2\eta \log n}. \tag{8}$$

Now define $G_r(w) = \langle w, x^{(r)} \rangle$. The vector $(G_1(w), \dots, G_{k^\star}(w))$ is jointly Gaussian with $\mathrm{Var}(G_r) = 1$ and $\mathrm{Cov}(G_r, G_s) = \langle x^{(r)}, x^{(s)} \rangle \le \rho$ for $r \ne s$. Applying Lemma B.3,

$$\sqrt{1-\rho}\, \mathbb{E}\Big[\max_{r \le k} Z_r \Big] \;\le\; \mathbb{E}_w M(T; w) \;\le\; \sqrt{1+\rho}\, \mathbb{E}\Big[\max_{r \le k} Z_r \Big]. \tag{9}$$

Combining (8) and (9), and using $S = \frac{1}{m} \sum_{j=1}^m M(\tilde{X}; w_j)$ with i.i.d. w_j, we obtain

$$L(k^\star) \;:=\; \sqrt{1-\rho}\, \mathbb{E}\Big[\max_{r \le k} Z_r \Big] - \sqrt{2\eta \log n} \;\le\; \mathbb{E}[S] \;\le\; \sqrt{1+\rho}\, \mathbb{E}\Big[\max_{r \le k} Z_r \Big] + \sqrt{2\eta \log n} \;=:\; U(k^\star). \tag{10}$$

Step 2: A uniform positive gap for geometric neighbors. Fix $t \in \{2, \dots, n\}$ and consider the two regimes $k^\star \le t$ versus $k^\star \ge t'$, where $t' = \lceil (1+\varepsilon)t \rceil$. Using monotonicity of G_k and (10),

$$\sup_{k^\star \le t} \mathbb{E}[S] \;\le\; U(t), \qquad \inf_{k^\star \ge t'} \mathbb{E}[S] \;\ge\; L(t').$$

Thus the *gap* between these regimes is at least

$$\mathrm{Gap}(t) \;:=\; L(t') - U(t) \;=\; \sqrt{1-\rho}\, \mathbb{E}\Big[\max_{r \le t'} Z_r \Big] - \sqrt{1+\rho}\, \mathbb{E}\Big[\max_{r \le t} Z_r \Big] - 2\sqrt{2\eta \log n}.$$

By Lemma B.6, under the parameter condition $\rho \le c_\rho \varepsilon / \ln n$ with c_ρ small enough, $\sqrt{1-\rho}\, \mathbb{E}\Big[\max_{r \le t'} Z_r \Big] - \sqrt{1+\rho}\, \mathbb{E}\Big[\max_{r \le t} Z_r \Big] = \Omega\big(\frac{\varepsilon}{\sqrt{\ln t}} \big)$.

Therefore, for all $t \le n$,

$$\mathrm{Gap}(t) \;\ge\; c_1 \frac{\varepsilon}{\sqrt{\ln n}} \;-\; 2\sqrt{2\eta \log n}.$$

Also, under the parameter condition $\eta \le c_\eta \varepsilon^2 / (\ln n)^2$ with c_η small enough, the negative term is of the same order as the leading term, so there exists an absolute $c > 0$ such that

$$\mathrm{Gap}(t) \;\ge\; c \frac{\varepsilon}{\sqrt{\ln n}} \qquad \text{for all } t \in \{2, \dots, n\}. \tag{11}$$

Step 3: Concentration and correctness of each threshold test. For each j, $M_j(\tilde{X}) = \max_i \langle w_j, \tilde{x}_i \rangle$ is a 1-Lipschitz function of w_j (since $\|\tilde{x}_i\| = 1$), so by Lemma B.4,

$$\Pr\Big(|S - \mathbb{E}S| \ge u \Big) \;\le\; 2 \exp\Big(-\frac{mu^2}{2} \Big).$$

Fix t and corresponding threshold $\theta = \frac{1}{2}(U(t) + L(t'))$. If $k^\star \le t$, then $\mathbb{E}S \le U(t) = \theta - \mathrm{Gap}(t)/2$, hence

$$\Pr(S > \theta) \;\le\; \Pr\Big(S - \mathbb{E}S > \mathrm{Gap}(t)/2 \Big) \;\le\; \exp\Big(-\frac{m\, \mathrm{Gap}(t)^2}{8} \Big).$$

If $k^\star \ge t'$, then $\mathbb{E}S \ge L(t') = \theta + \mathrm{Gap}(t)/2$, hence similarly

$$\Pr(S \le \theta) \;\le\; \exp\Big(-\frac{m\, \mathrm{Gap}(t)^2}{8} \Big).$$

Thus each test fails with probability at most $2\exp(-m\mathrm{Gap}(t)^2/8)$.

Step 4: Union bound over the geometric grid. We run at most $R = O(\frac{\ln n}{\varepsilon})$ thresholds. By (11), $\mathrm{Gap}(t)^2 \geq c^2\varepsilon^2/\ln n$. So choosing

$$m \;\geq\; C \cdot \frac{\ln n}{\varepsilon^2} \cdot \ln\Big(\frac{2R}{\delta}\Big)$$

ensures each test fails with probability at most δ/R, and by a union bound all tests succeed simultaneously with probability at least $1-\delta$.

Step 5: Conclude multiplicative accuracy. On the event all tests succeed, the algorithm returns the first r with $S \leq \theta_r$. By correctness of the tests, this implies $k^\star \leq t_{r+1} = \hat{k}$. Also, for the previous index $r-1$ (if it exists), we must have $S > \theta_{r-1}$, implying $k^\star > t_{r-1}$. Since $t_{r+1} \leq (1+\varepsilon)t_r \leq (1+\varepsilon)k^\star$ by construction of the geometric grid, we conclude $\hat{k} \leq (1+\varepsilon)k^\star$. □

B.3 Corollary: Existence of deterministic weights

Corollary B.8 (Deterministic max-aggregator weights)**.** *Under the conditions of Theorem 4.4, for any distribution $\mathcal{D}$ over $(\eta,\rho)-$clusterable sequences there exist fixed $w_1,\ldots,w_m$ such that MaxSketch achieves $(1+\varepsilon)$ multiplicative approximation with probability at least $1-\delta$ over a fresh draw $X \sim \mathcal{D}$.*

Proof. Let $\mathcal{D}$ be a distribution over (η,ρ)-clusterable sequences and let $X \sim \mathcal{D}$ be a sequence with an unknown number of centers $k^\star$.

Let

$$W = (w_1,\ldots,w_m), \qquad w_j \sim \mathcal{N}(0, I_d) \text{ independently.}$$

Define the estimator $\hat{k}(X;W) := t_{r+1}$, where $r = \min\{r : S(X;W) \leq \theta_r\}$ with $S(X;W) := \frac{1}{m}\sum_{j=1}^m \max\langle w_j, \tilde{x}_i\rangle$ and the thresholds θ_r defined as Theorem B.7. If no such r exists, set $\hat{k}(X;W) := t_R = n$.

Now we define the failure indicator

$$F(X,W) := \mathbb{1}\left\{\hat{k}(X;W) \notin [k^*, (1+\varepsilon)k^*]\right\} \tag{12}$$

From Theorem B.7, for any (η,ρ)-clusterable sequence X, the probability of failure with respect to the Gaussian weights W is bounded by

$$\text{for all } (\eta,\rho)-\text{clusterable sequences } X, \quad \Pr_W(F(X,W)=1) \leq \delta.$$

Moreover, we can take the *expectation over the randomness of $X \sim \mathcal{D}$* to bound the average failure probability across sequences. That is, defining

$$g(W) := \mathbb{E}_X[F(X,W)] = \Pr_{X\sim\mathcal{D}}\big[\hat{k}(X;W) \notin [k^\star, (1+\varepsilon)k^\star]\big],$$

using the linearity of expectation we define the average probability of failure:

$$\mathbb{E}_W[g(W)] = \mathbb{E}_W\mathbb{E}_X[F(X,W)] = \mathbb{E}_X \Pr_W[F(X,W)=1] \leq \delta. \tag{13}$$

Since $\mathbb{E}_W[g(W)] \leq \delta$, by the probabilistic method there exists a deterministic $W^\star$ such that $g(W^\star) \leq \delta$, i.e., $\Pr_{X\sim\mathcal{D}}[\hat{k}(X;W^\star) \notin [k^\star,(1+\varepsilon)k^\star]] \leq \delta$, which completes the proof. □

B.4 Theorem 2: Efficiently finding weights from count-only data

Theorem B.9 (Theorem 4.7: Learning a max-aggregator readout)**.** *Under the conditions of Theorem 4.4 there exists a polynomial-time learning algorithm which, given $N = \mathrm{poly}(\ln n, 1/\varepsilon, \ln(1/\delta))$ i.i.d. training samples $(X_1, k_1), \ldots, (X_N, k_N) \sim \mathcal{D}$, outputs a monotone readout function $f : \mathbb{R} \to \mathbb{R}$ such that, for a fresh test sample $(X, k) \sim \mathcal{D}$,*

$$k \;\le\; f(S(X)) \;\le\; (1+\varepsilon)k$$

with probability at least $1 - \delta$.

Proof. For fixed Gaussian weights $w_1, \ldots, w_m$, define

$$G_k \;:=\; \mathbb{E}\big[S(X) \mid X \text{ has exactly } k \text{ centers}\big].$$

Step 1: Concentration of the sketch. For any fixed sequence X, the map $w \mapsto \max_i \langle w, x_i \rangle$ is 1-Lipschitz. By Gaussian concentration,

$$\Pr\big(|S(X) - \mathbb{E}S(X)| \ge t\big) \;\le\; 2\exp\Big(-\frac{mt^2}{2}\Big).$$

Let

$$\Delta := c_0 \frac{\varepsilon}{\sqrt{\ln n}},$$

where $c_0 > 0$ is the constant from Lemma B.6. Fix $\zeta \in (0,1)$. If

$$m \;\ge\; C_0 \cdot \frac{\ln n}{\varepsilon^2} \cdot \ln\Big(\frac{1}{\zeta}\Big),$$

then by integrating the above tail bound over $X \sim \mathcal{D}_k$ and applying Markov's inequality, we obtain: with probability at least $1 - \zeta$ over the draw of the Gaussian weights,

$$\Pr_{(X,k)\sim\mathcal{D}}\Big(|S(X) - G_k| \ge \tfrac{\Delta}{4}\Big) \;\le\; \zeta.$$

Condition on this event for the remainder of the proof.

Step 2: Monotonicity and separation. By Lemmas B.2, B.3, and B.6, the function $k \mapsto G_k$ is nondecreasing and satisfies the uniform gap

$$G_{(1+\varepsilon)k} - G_k \;\ge\; \Delta \qquad \text{for all } k \ge 2.$$

Step 3: Discretization and oracle monotone readout. Define multiplicative levels

$$L_0 = 1, \qquad L_{t+1} = \lceil (1+\varepsilon) L_t \rceil,$$

and let $T = \min\{t : L_t \ge n\}$. Note $T = O((1/\varepsilon)\ln n)$.

For each $t = 0, \ldots, T-1$, choose a threshold

$$\tau_t \in \big(G_{L_t} + \tfrac{\Delta}{4},\, G_{L_{t+1}} - \tfrac{\Delta}{4}\big),$$

which is nonempty by Step 2. Define the oracle monotone function

$$f^\star(s) \;:=\; L_{t+1} \quad \text{if } s \in [\tau_t, \tau_{t+1}).$$

By construction and the concentration event of Step 1,

$$\Pr_{(X,k)\sim\mathcal{D}}\big(f^\star(S(X)) \notin [k, (1+\varepsilon)k]\big) \;\le\; O(\zeta).$$

Step 4: Learning the monotone readout. By Step 3, there exists an oracle monotone step function $f^\star$ with at most T breakpoints $\tau_0^\star < \cdots < \tau_{T-1}^\star$ and range contained in $\{L_1, \ldots, L_T\}$, satisfying

$$\Pr_{(X,k)\sim\mathcal{D}}\big(f^\star(S(X)) \notin [k, (1+\varepsilon)k]\big) \;\le\; O(\zeta).$$

Since $f^\star$ is determined by its T breakpoints, learning $f^\star$ from data reduces to learning T one-dimensional thresholds. Each such threshold is a binary classifier on $\mathbb{R}$ and the class of one-dimensional threshold functions has VC dimension 1. Standard uniform convergence bounds for VC classes (see, e.g., [25]) then imply that each breakpoint can be learned, from

$$N_0 \;=\; O\Big(\tfrac{1}{\zeta}\ln\tfrac{T}{\delta}\Big)$$

i.i.d. samples, with excess population error at most ζ and probability of failure at most δ/T. A union bound over the T breakpoints implies that, with probability at least $1-\delta$ over the training sample, *all* learned breakpoints $\hat{\tau}_0, \ldots, \hat{\tau}_{T-1}$ are simultaneously ζ-accurate.

Let $\hat{f}$ be the resulting monotone step function. Because $f^\star$ was constructed so that its output always lies in the correct multiplicative interval (Step 3), and $\hat{f}$ approximates $f^\star$ in population loss up to $O(\zeta)$, the correctness guarantee transfers:

$$\Pr_{(X,k)\sim\mathcal{D}}\Big(\hat{f}(S(X)) \notin [k, (1+\varepsilon)k]\Big) \;\le\; O(\zeta).$$

The total sample complexity is $N = T \cdot N_0 = \mathrm{poly}(\ln n,\, 1/\varepsilon,\, \ln(1/\delta))$.

Step 5: Parameter choice and conclusion. Recall that the number of multiplicative levels satisfies $T = O\big(\frac{1}{\varepsilon}\ln n\big)$. Set $\zeta = \Theta\big(\frac{\delta}{T}\big)$. With this choice, a union bound over the T level transitions ensures that the total failure probability is at most δ.

Substituting this value of ζ into the concentration requirement of Step 1 yields the sketch dimension

$$m \;\ge\; C \cdot \frac{\ln n}{\varepsilon^2}\,\ln\Big(\frac{\ln n}{\varepsilon\delta}\Big),$$

while the sample complexity in Step 4 becomes

$$N = \mathrm{poly}\big(\ln n,\, 1/\varepsilon,\, \ln(1/\delta)\big).$$

Taking a union bound over the randomness of the Gaussian weights, the training sample, and the test sample completes the proof.

□

C Experimental Details

C.1 MNIST

Dataset and preprocessing. We use the standard MNIST dataset consisting of 60,000 training images and 10,000 test images. All images are scaled to the range $[0, 1]$ and standardized using the training set mean and standard deviation. No data augmentation is applied.

Encoder. We use a small convolutional neural network composed of two convolutional blocks. The first block applies a 3×3 convolution with 32 channels, followed by batch normalization, a ReLU activation, and 2×2 max pooling. The second block applies a 3×3 convolution with 64 channels, again followed by batch normalization, ReLU, and 2×2 max pooling. The resulting feature maps are flattened and passed through a linear layer producing an embedding of dimension $d = 256$. All embeddings are L2-normalized before being passed to the aggregation module.

Aggregators. We evaluate three aggregation mechanisms. MaxSketch uses $m = 4{,}096$ fixed random Gaussian projections and computes the maximum projection along each direction. These statistics are then passed through a two-layer MLP with hidden width 256 and ReLU activation to produce the final count prediction. DeepSets applies a two-layer MLP with hidden width 256 and output width 256 with ReLU activation to each embedding individually, aggregates representations via summation, and then applies a final two-layer MLP with hidden width 256 and ReLU activation to predict the count. The LSTM baseline processes the sequence of embeddings using a single-layer LSTM with hidden size 256, and the final hidden state is mapped to the count via a linear layer.

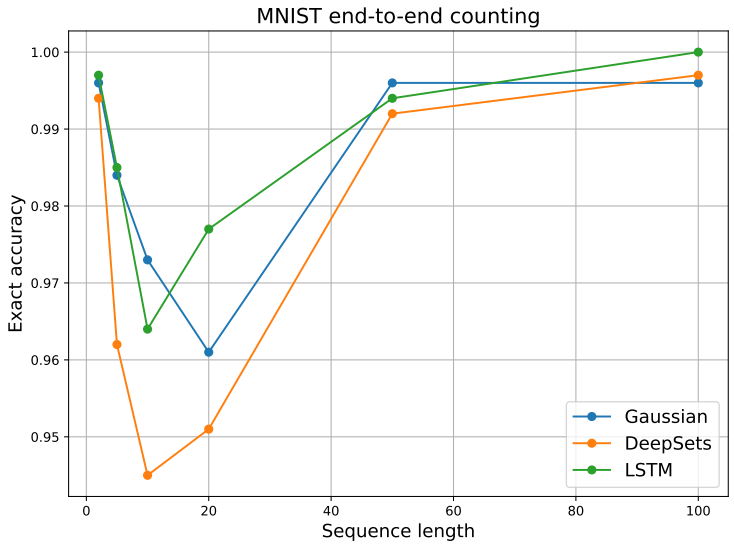


Figure 2: Exact counting accuracy of three baseline models, MaxSketch, DeepSets, and LSTM, on MNIST sequences of varying lengths. All models achieve near-perfect accuracy across all sequence lengths, and the ± 1 accuracy reaches 100%, indicating that predictions are at most one count away from the true number of digits.

Training. All models are trained using the Adam optimizer with learning rate 0.001 and batch size 32 sequences. The loss function is mean squared error between predicted and true counts. Training is performed for 50 epochs, with evaluation after each epoch.

Stream construction. Each training sequence is generated by first sampling $k \sim \mathrm{Uniform}\{1, \ldots, 10\}$. We then sample k classes uniformly without replacement, and construct a sequence of length n by repeatedly sampling a class uniformly from the chosen set and then sampling an image uniformly from that class. Each sequence is resampled independently at every epoch. Test sequences are constructed using the same procedure but with test images.

Experiments. We evaluate accuracy across sequence lengths by training and testing at each $n \in \{2, 5, 10, 20, 50, 100\}$ independently. For extrapolation, MaxSketch is trained at $n_{\text{train}} \in \{10, 20, 50\}$ and tested at $n_{\text{test}} \in \{150, 250, 500\}$. In addition, we compute 1-nearest-neighbour accuracy using cosine distance to the nearest training embedding.

C.2 CIFAR-10

Dataset and preprocessing. We use the standard CIFAR-10 dataset with 50,000 training images and 10,000 test images. Images are normalized using channel-wise statistics computed on the training set. During pretraining, we apply random cropping with padding 4 and random horizontal flipping. No augmentation is used during count training or evaluation.

Encoder. The encoder is a convolutional neural network composed of four convolutional blocks. Each block consists of a 3×3 convolution with C channels, followed by batch normalization, ReLU activation, and 2×2 max pooling. Channel widths are given by 64. The resulting features are flattened and projected into an embedding space of dimension $d = 256$, followed by L2 normalization.

Aggregators. We use the same aggregation architectures as in MNIST, namely MaxSketch, DeepSets, and LSTM, with identical configurations except where otherwise specified.

Training regimes. We evaluate three regimes. In the supervised frozen regime, the encoder is pretrained on CIFAR-10 using cross-entropy loss for 50 epochs, achieving a test accuracy of 91%, and is then frozen during count training. Only the readout module is trained. In the supervised

fine-tuning regime, the pretrained encoder is further fine-tuned during count training using learning rate 0.001, while the readout is trained simultaneously with learning rate 0.01. In the end-to-end regime, the encoder is initialized randomly and trained jointly with the readout using only count supervision, without access to class labels.

Training. All regimes use the Adam optimizer and mean squared error loss. Training is performed with batch size 32 sequences for 50 epochs.

Stream construction and experiments. Stream construction follows the same procedure as MNIST, applied to CIFAR-10 classes and images. Models are trained at $n_{\text{train}} \in \{10, 20, 50\}$ and evaluated at the same for each regime, as reported in Table 2.

C.3 Faces (LFW + CelebA + Multi-PIE)

Datasets. We construct a unified face dataset by merging three benchmarks: LFW, CelebA, and CMU Multi-PIE. From each dataset we retain identities with at least two images. The resulting identity pool contains approximately 10,000 identities, which are split uniformly at random by identity (not by image) into a calibration set and an evaluation set of equal size (5,103 identities each).

Embeddings. We use the DeepFace [24] implementation of the VGG-Face model to extract 4,096-dimensional embeddings. The model uses its default detector and alignment pipeline. All embeddings are L2-normalized and the encoder is kept frozen throughout all experiments.

MaxSketch configuration. MaxSketch uses $m = 4{,}096$ fixed random Gaussian projections drawn from $\mathcal{N}(0, I_{4096})$. For a stream X, we compute the statistic $S(X) = \frac{1}{m}\sum_{j=1}^{m} \max_i \langle w_j, x_i \rangle$. The readout function is a one-dimensional isotonic regression model trained on the calibration split using scikit-learn's IsotonicRegression. Calibration streams are generated using identities from the calibration set with k sampled uniformly from the evaluation range.

Stream construction. Each evaluation stream is constructed by first sampling a target value $k \in \{100, 200, \ldots, 5100\}$. We then sample k identities uniformly without replacement from the evaluation set. A stream of length $n = 20{,}000$ is generated by repeatedly sampling an identity uniformly from this set and then sampling an image uniformly from that identity. For each value of k, we generate 100 independent streams.

Zhang baseline. We evaluate the robust F_0-estimation algorithm of Zhang [30], corresponding to Algorithm 1 in metric spaces. The distance threshold parameter α is swept over a predefined range $\{0.2, 0.21, 0.22, \ldots, 0.8\}$, and the optimal value $\alpha^* = 0.40$ is selected based on minimum mean absolute error on the calibration set. All distances are computed using Euclidean distance on L2-normalized embeddings. Remaining hyperparameters follow the original implementation.

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